CIA-1

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1(a) Caesar Cipher

Given

Key = 7

PlainText = MEETATNOON

Caesar Cipher Encryption => $C=(P+K) \bmod 26$

Caesar Cipher Decryption => $P=(C-K) \bmod 26$

PlainText	Number	key	calculation	Cipher Text
M	12	7	(12+7) mod 26 ⇒ 19	T
E	4	7	(4+7) mod 26 ⇒ 11	L
E	4	7	(4+7) mod 26 ⇒ 11	L
T	19	7	(19+7) mod 26 ⇒ 0	A
A	0	7	(0+7) mod 26 ⇒ 7	H
T	19	7	(19+7) mod 26 ⇒ 0	A
N	13	7	(13+7) mod 26 ⇒ 20	U
O	14	7	(14+7) mod 26 ⇒ 21	V
O	14	7	(14+7) mod 26 ⇒ 21	V
N	13	7	(13+7) mod 26 ⇒ 20	U

Cipher Text produced : TLLAHAUVVU

Ranks

A - 0
B - 1
C - 2
D - 3
E - 4
F - 5
G - 6
H - 7
I - 8
J - 9
K - 10
L - 11
M - 12
N - 13
O - 14
P - 15
Q - 16
R - 17
S - 18
T - 19
U - 20
V - 21
W - 22
X - 23
Y - 24
Z - 25

1 Caesar Cipher Decryption

$P = (c - k) \bmod 26$

Cipher Text	key	Calculation	Plain Text
T (19)	7	(19-7) mod 26 ⇒ 12	M
L (11)	7	(11-7) mod 26 ⇒ 4	E
L (11)	7	(11-7) mod 26 ⇒ 4	E
A (0)	7	(0-7) mod 26 ⇒ 19	T
H (7)	7	(7-7) mod 26 ⇒ 0	A
A (0)	7	(0-7) mod 26 ⇒ 19	T
U (20)	7	(20-7) mod 26 ⇒ 13	N
V (21)	7	(21-7) mod 26 ⇒ 14	O
V (21)	7	(21-7) mod 26 ⇒ 14	O
U (20)	7	(20-7) mod 26 ⇒ 13	N

Plain Text produced is MEETATNOON

1(b) Vigenere Cipher

PlainText = MEETATNOON

key = SUN

Vigenere Cipher Encryption $C = P + K \bmod 26$

Vigenere cipher decryption ⇒ $P = (C - K) \bmod 26$

Plain Text	Key	Calculation	Cipher Text
M (12)	S (18)	(12+18) mod26 ⇒ 4	E
E (4)	Y (24)	(4+24) mod26 ⇒ 24	Y
G (6)	N (13)	(6+13) mod26 ⇒ 17	R
T (19)	S (18)	(19+18) mod26 ⇒ 11	L
A (0)	Y (24)	(0+24) mod26 ⇒ 24	Y
T (19)	N (13)	(19+13) mod26 ⇒ 6	G
N (13)	S (18)	(13+18) mod26 ⇒ 5	F
O (14)	Y (24)	(14+24) mod26 ⇒ 8	I
O (14)	N (13)	(14+13) mod26 ⇒ 1	A
N (13)	S (18)	(13+18) mod26 ⇒ 5	F

Cipher Text: EYRLYGFIAF

Vigenere cipher decryption: $P = (C-K) \bmod 26$

CipherText	Key	Calculation	Plain Text
E (4)	S (18)	(4-18) mod26 ⇒ 12	M
Y (24)	Y (24)	(24-24) mod26 ⇒ 4	E
R (17)	N (13)	(17-13) mod26 ⇒ 4	E
L (11)	S (18)	(11-18) mod26 ⇒ 19	T
Y (24)	Y (24)	(24-24) mod26 ⇒ 0	A
G (6)	N (13)	(6-13) mod26 ⇒ 19	T
F (5)	S (18)	(5-18) mod26 ⇒ 13	N
I (8)	Y (24)	(8-24) mod26 ⇒ 14	O
A (1)	N (13)	(1-13) mod26 ⇒ 14	O
F (5)	S (18)	(5-18) mod26 ⇒ 13	N

Plain Text received is : MEETATNOON

(1) Comparison b/w Caesar Cipher & Vigenere Cipher:

* Both methods uses key to encrypt & decrypt.
* Same key used for both Encryption & Decryption
* In Caesar cipher it uses same fixed key value for Both Encryption & Decryption.

(2) Given,

PlainText : MATH

$$\text{Key} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

MATH

Here,

M = 12

A = 0

T = 19

H = 7

$P_1 = MA \Rightarrow \begin{bmatrix} 12 \\ 0 \end{bmatrix}$

$P_2 \Rightarrow TH \Rightarrow \begin{bmatrix} 19 \\ 7 \end{bmatrix}$

$K^{-1} = \frac{1}{|K|}$

$\Rightarrow \frac{1}{(ad-bc)} \Rightarrow \frac{1}{(21-10)} \Rightarrow \frac{1}{11}$

156

$K^{-1} \Rightarrow \frac{1}{11}\begin{bmatrix} -7 & 2 \\ -5 & 3 \end{bmatrix}$

$\Rightarrow 19\begin{bmatrix} -7 & 2 \\ -5 & 3 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} -133 & 38 \\ -95 & 57 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 3 & 18 \\ 14 & 5 \end{bmatrix} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$

$P \Rightarrow K^{-1}C \bmod 26$

$\Rightarrow \begin{bmatrix} 20 \\ 5 \end{bmatrix} \begin{bmatrix} 101 \\ 14 \end{bmatrix} \bmod 26$

$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix} \bmod 26$

398
8
30
108
168
230
398
8
4

$\Rightarrow \begin{bmatrix} 142 \\ 130 \end{bmatrix} \bmod 26$

$\Rightarrow 12$
0

$\Rightarrow 12 = M, \ 0 = A$

Encryption $C = KP \bmod 26$

① $C = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} \bmod 26$

$\Rightarrow \begin{bmatrix} 36 \\ 60 \end{bmatrix} \bmod 26$

$\Rightarrow \begin{bmatrix} 10 \\ 8 \end{bmatrix}$

$0 = K$

$8 \Rightarrow I$

② $C = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 0 \end{bmatrix} \bmod 26$

$\Rightarrow \begin{bmatrix} 91 \\ 144 \end{bmatrix} \bmod 26$

$\Rightarrow \begin{bmatrix} 19 \\ 14 \end{bmatrix}$

$19 = T$

$14 \Rightarrow O$

Decryption $P = K^{-1} C \bmod 26$

$K^{-1} = \frac{1}{|K|} K^{*}$

$K^{*} = \begin{bmatrix} -d & b \\ -c & a \end{bmatrix} \Rightarrow \begin{bmatrix} -7 & 2 \\ -5 & 3 \end{bmatrix}$

P = $\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$ $\begin{bmatrix} 19 \\ 14 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 19 \\ 7 \end{bmatrix}$

$\Rightarrow 19 = 7 \quad H = 7,$

(3) Given:

200 clinics have to send 1GB of MRI files to servers.

Comparision between AES-256 and RSA and Explain which is more suitable

(a)

Feature	AES-256	RSA
Purpose	Data Encryption	key Encryption ;
Data	large files	small files
Speed	very fast	slow
key	a secret key	Public key & a secret key
Type	Symmetric	Asymmetric

Key Usage:

* AES-256 uses a single secret key for file Transfer.

* RSA has both a public key and a private key.

* Public key is shared where as the private key is not shared.

(ii) Performance

* AES-256 can handle large amount of data at once.

* Where as the RSA cannot handle the large amount of files.

(iii) Sustainability

for 200 clinics pairs of AES

Keys = $\frac{n(n-1)}{2}$

$\Rightarrow \frac{200(200-1)}{2}$

$\Rightarrow \frac{200(199)}{2}$

$\Rightarrow$

(iv) For RSA

* for each clinic it contains a public key and a private key pair

* and a pair at the sever

Total = 201 pairs

* Digital signatures are used to verify public key.

Result :

AES-256 is has better performance at encrypting larger amount of files whereas RSA is very good at protecting the files.

(5) Hybrid Encryption workflow

(i) First create a AES-256 public key

(ii) Encrypt the MRI files into AES-256 key

(iii) encrypt the AES-256 with RSA public key

(iv) send it to the receiver

(v) the receiver decrypts the RSA using private key.

(vi) He gets the MRI files securely then.

(6) Secure key storage and Rotation

* stores keys securely

* Access only authorised users to Access keys

(5) Introduction

The check the types of encryption with encryption techniques used a suitable for an problem. using some key value.

The types of encryption techniques used are

(i) Caesar cipher

(ii) Vigenere cipher

(iii) AES-256

(iv) RSA

(a) Classical Encryption Techniques Used

(i) Caesar Cipher

Its Encryption formula $c=(p+x) \bmod 26$

key = 3 , plain text = HELLO

Plain Text	key	calculation	Cipher Text
7	3	$(7+3) \bmod 26 \Rightarrow 10$	
4	3	$(4+3) \bmod 26 \Rightarrow 7$	
11	3	$(11+3) \bmod 26 \Rightarrow 14$	
11	3	$(11+3) \bmod 26 \Rightarrow 14$	
14	3	$(14+3) \bmod 26 \Rightarrow 17$	

Caesar cipher

It's encryption formula $c=(p+k) \bmod 26$

Ex. plaintext = MEETATNOON

key = 7

plaintext	Number	key	addition	ciphertext
M	12	7	19	T
E	4	7	11	L
E	4	7	11	L
T	19	7	0	A
A	0	7	7	H
T	19	7	0	A
N	13	7	20	U
O	14	7	21	V
O	14	7	21	V
N	13	7	20	U

cipher Text = TLLAHAUVVU

Now, do decryption using $P=(C-K) \bmod 26$ we get

MEETATNOON as plaintext.

(ii) Vigenere cipher

PlainText : MEETATNOON

key = SUN.

plainText	key	addition	cipher text
M (12)	S	4	E
E (4)	U	24	Y
E (4)	N	13	R
T (19)	S	1	L
A (0)	U	20	U
T (19)	N	6	G
N (13)	S	5	F
O (14)	U	8	I
O (14)	N	-1	A
N (13)	S	5	F

we get cipher text : EYRLUGFIAF

If we do decryption using $D = (C - k) \bmod 26$ we get the MEETATNOON as plain Text.

(5) Modern Encryption Techniques used

① AES-256

* It is a Data Encryption Technique
* It is used to encrypt large files & Data
* It is a very fast Encryption Technique
* It uses a secret key only.
* It is a symmetric type.

key usage of It:

* AES-256 It uses a single key for file transfer.
* It can transfer large amount of MRI files or other files at once.
* It is a very sustainable key.
* It is mainly used to transfer longer files.

(ii) RSA

* It is a key Encryption Technique
* It is used to transfer small files
* It is very slow
* It has a public & private key pair
* It is Asymmetric value

Key Usage.

* RSA has both a public key & a private key.
* Public key is shared & whose as the private key is not shared.

Comparison and Conclusion:

Among these classic & Modern Encryption Techniques AES-256 is considered as the Best Encryption Technique. Due to its huge files handles value. It is Symmetric & mainly it is very fast.

Given

(1)

p=11, q=19, e=7, M=5

p=11, q=19

Public modulus

n = p×q

= 11×19

n=209

Euler's Totient

$\phi(n) = (p-1)(q-1)$

$= (11-1)(19-1)$

$= 10 \times 18$

$\phi(n) = 180$

Public key Exponent (e) = 7

GCD(7, 180) = 1

Find private key (d)

$ed \equiv 1 \mod \phi(n)$

$7d \equiv 1 \mod 180$

Using Euclidean Algorithm

$180 = 7 \times 25 + 5$

$7 = 5 \times 1 + 2$

$5 = 2 \times 2 + 1$

Back substitution:

$1 = 5 - 2(2)$

$= 5 - 2(7-5)$

$= 3(5) - 2(7)$

$= 3(180 - 25 \times 7) - 2(7)$

$= 3(180) - 77(7)$

$d = -77 \bmod 180$

$\boxed{d = 103}$

Public key, $(e, n) = (7, 209)$

Private key: $(d, n) = (103, 209)$

Encryption.

$M = 5$

$\boxed{C = M^e \bmod n}$

$C = 5^7 \bmod 209$

$5^7 = 5^4 \times 5 \times 5$

$C \Rightarrow 207 \times 25 \times 5 \bmod 209$

$\boxed{C = 168}$

$5^3 = 25$

$5^4 \Rightarrow 625$

$625 \bmod 209$

$= 207$

decryption

$M = C^d \bmod n$

$M = 168^{103} \bmod 209$

using chinese remainder theorem

modulo 11

$168 \bmod 11 \Rightarrow$

$= 3$

$103 \bmod 10$

$= 3$

$3^3 \bmod 11 = 27 \bmod 11 = 5$

modulo 19

$168 \bmod 19$

$= 16$

$103 \bmod 18$

$= 13$

$16^{13} \bmod 19 = 5$

from modulo 11 & 19

$M = 5$